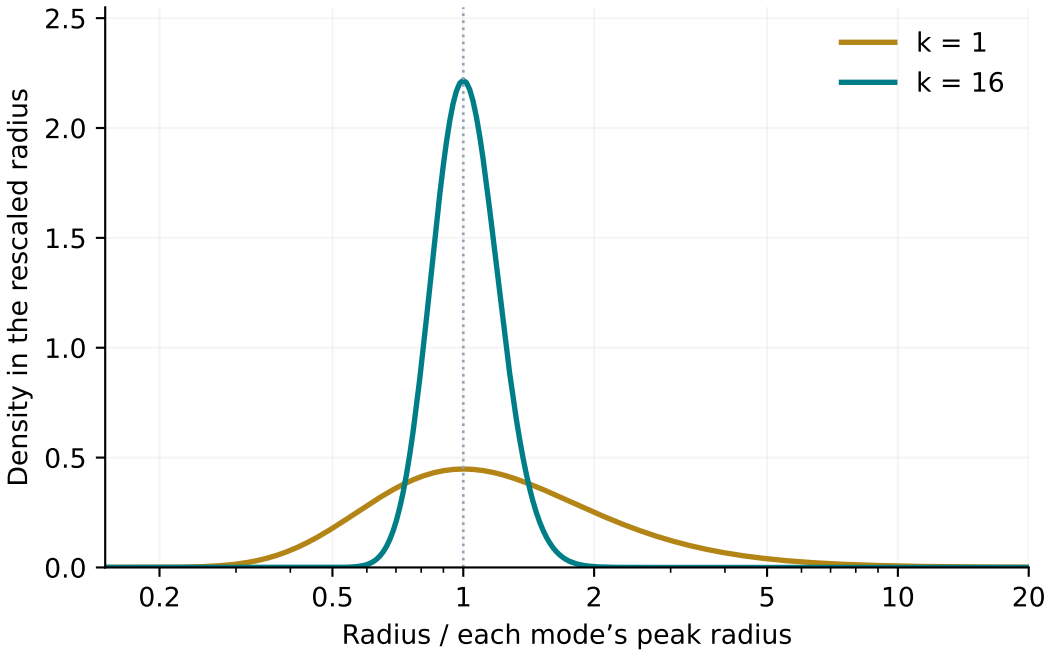
Spinful transverse modes and a computed two-dimensional quantum packet

Exact transverse benchmark + short local evolution | Neither panel gives a return or escape probability

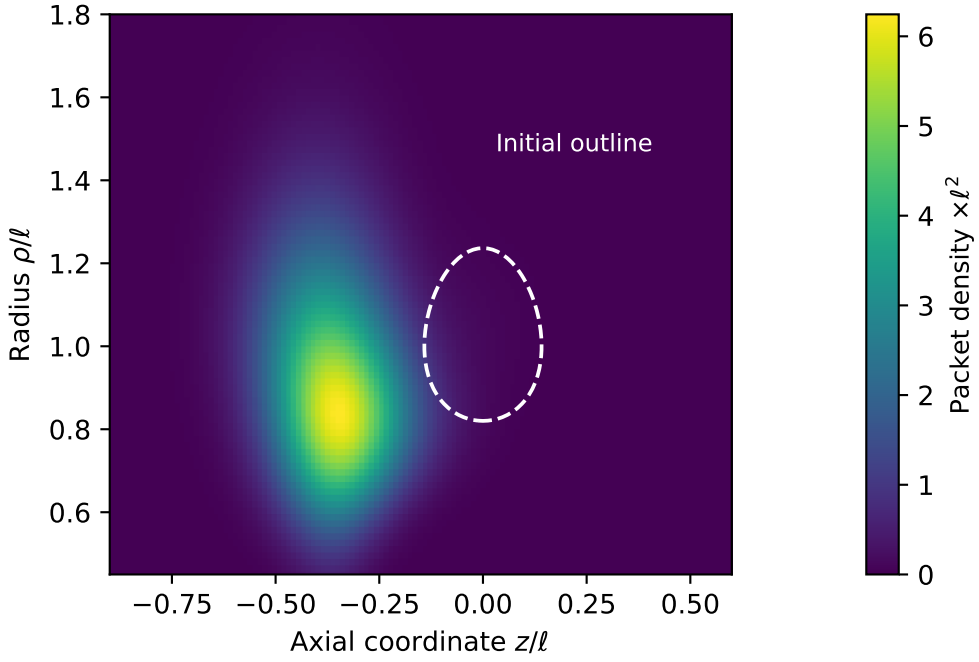
A. Exact transverse radial modes



Each full curve is normalized; radii are scaled separately.

The k = 1 mode has a long radial tail.

B. Evolved packet at t = 0.4 ℓ/c



Dashed: initial half-maximum density; local view shown.

$\mu = 16.5$, $\ell = qd/(2\mu)$; includes both spinor components.

The k = 0 slice mode is not normalizable. This does not determine a three-dimensional escape probability.

The packet evolves briefly in a periodic box; no mouth-to-mouth or outgoing-scattering conditions are imposed.

No constructed stable medium or 13 Sep 2027 → 12 Sep 2027 passage demonstrated.